CCCA Cell-State Analysis — Raghavan et al. 2021 (PDAC)

GO / NO-GO GATES

Gate 1 — integration & continuity: GO

21 samples intermix per-neighbourhood (iLISI 1.0→7.0 of 21) on one continuous component (0 islands). Developmental axis orthogonal to proliferation (|rho|≤0.15 vs cell cycle) → whole manifold kept.

Gate 2 — module geography: GO (non-branching)

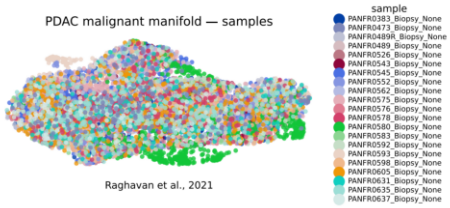
Single Gast/pSom→Crest axis: Gast & pSom co-vary (r=+0.25), both oppose Crest (r=-0.75 / -0.65); Crest an inverse terminal gradient. No Gast-vs-pSom arm split. Crest peaks at the terminus, not the Gast arm → no inversion flag.

Gate 3 — hub cross-repression: NA

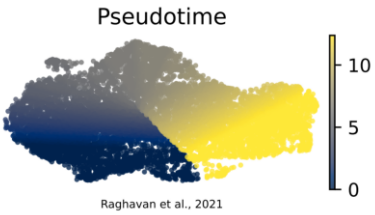
No bifurcation (PPT tree tips share one pSom-high identity; branch-node count unstable 2–4). Branching-only gate does not apply; 04_correlation / 05_modules not produced.

STEP FIGURES

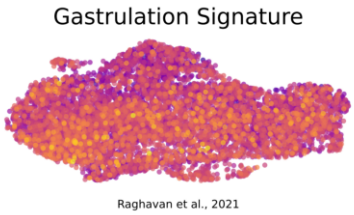
Gate 1: sample integration



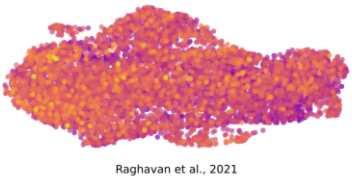
Trajectory / pseudotime



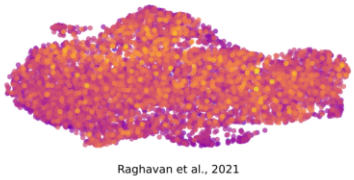
Gastrulation



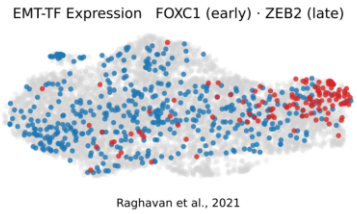
Somitogenesis



Neural Crest



EMT-TFs: FOXC1→ZEB2



Dataset. Metastatic pancreatic ductal adenocarcinoma, Seq-Well; 7,830 malignant cells, 21 samples (mostly liver mets).

Verdict. The embryonic pattern reproduces in collapsed form: one continuous gastrulation/somitogenesis → neural-crest axis (global means Gast -0.011 / pSom +0.072 / Crest -0.024; Linear-pSom/Crest), rooted at the Gast/pSom-high origin, EMT-TFs FOXC1 (root) → ZEB2 (terminus). Non-branching — no competing arms, so Gate 3 is NA.

Parameter choices

Sample threshold	geom-mean across largest count gap (120→45) = 73 → 21/23 samples
Genes / PCA	overdispersed selection; PCA 50, n_pcs = 20 (43% var; Puram-40/Kurten-20 scale)
Integration	Harmony on sample, theta = 10 (mixing plateaus; strongest without collapse)
UMAP / neighbours	n_neighbors = 30, spread = 1.3; island removal = radius-graph (0 removed)
Geometry	principal curve (ElPiGraph), Nodes 20, lambda 0.01, mu 50; fold-free (0 forks)
Root	derived: Gast/pSom-high, Crest-low tip (developmental origin; Crest is late)
EMT-TFs	detection-gated wide panel → FOXC1 (early) · ZEB2 (late)